AuralGuard: Multi-View One-Class Learning for Generalizable Detection of AI-Generated Speech in the Wild

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***Abstract*—Modern text-to-speech and voice-conversion systems produce speech indistinguishable from real recordings, yet state-of-the-art spoofing detectors degrade sharply on *unseen* synthesizers and *real-world* channels. We present AuralGuard, a multi-view detector that fuses a self-supervised speech front-end (with learnable layer attention), a hand-crafted vocoder-artifact branch, and a spectro-temporal graph-attention back-end, trained with a one-class objective and a channel-robustness augmentation curriculum. On a cross-corpus benchmark spanning ASVspoof 2019/2021, In-the-Wild, WaveFake, and multilingual MLAAD, AuralGuard reduces zero-shot equal error rate by XX% relative to a strong SSL–AASIST baseline while remaining well-calibrated. We release code, weights, a live inference API, and a demo interface.**

***Index Terms*—audio deepfake detection, anti-spoofing, synthetic speech, self-supervised learning, one-class classification, generalization.**

I. Introduction

Recent advances in text-to-speech (TTS) and voice conversion (VC)—driven by large-scale neural architectures such as VALL-E, StyleTTS 2, and neural codec language models—now produce speech that human listeners and simple classifiers cannot reliably distinguish from genuine recordings [1]. While this technology enables beneficial applications in accessibility and creative production, it also introduces serious threats: voice-cloning fraud, disinformation through fabricated audio of public figures, and evasion of voice biometric authentication. The anti-spoofing community has responded with increasingly sophisticated detection models, and *in-domain* performance on benchmarks such as ASVspoof 2019 Logical Access (LA) now approaches 1% equal error rate (EER) [2], [3].

Despite this progress, three failure modes prevent reliable deployment in the wild. **G1—Generalization to unseen generators:** A detector trained on a fixed set of TTS/VC algorithms sees EER degrade to 5–30% when evaluated on synthesizers absent from its training corpus. Because new speech generation models appear monthly, a practical detector must generalize without retraining. **G2—Channel robustness:** Real-world audio reaches detection systems after compression (MP3, Opus, AAC), telephony coding, reverberation, and additive noise. The interaction of these distortions with spoofing artifacts is poorly understood, and most evaluations to date use clean, studio-quality conditions. **G3—Calibration and explainability:** Binary classification accuracy alone is insufficient for forensic or legal applications, which require well-calibrated confidence estimates and interpretable evidence for each decision. Prior work has largely treated calibration and explainability as afterthoughts.

The field thus needs a detection framework that simultaneously addresses open-set generalization, channel robustness, and calibrated decision-making. We reframe the task from closed-set binary classification to open-set, channel-robust, calibrated detection of AI-generated speech, and propose **AuralGuard** to address all three gaps.

AuralGuard is a multi-view architecture with four key design decisions. First, it fuses two complementary views of the input: *View A*, a self-supervised speech representation (WavLM-Large) with learnable per-layer attention that flexibly selects the most informative hidden states for deepfake detection; and *View B*, a hand-crafted feature stream that explicitly captures low-level vocoder artifacts through LFCC, modified group-delay, and CQT-phase coefficients features that SSL models may overlook. Second, a gated cross-attention fusion mechanism adaptively combines the two views before a spectro-temporal graph attention back-end (AASIST) produces an utterance-level embedding. Third, a one-class contrastive objective combining OC-Softmax [4] and supervised contrastive loss [5] learns a compact decision boundary around bona-fide speech, naturally rejecting unseen spoofs as out-of-distribution. Fourth, a channel-robustness augmentation curriculum—including RawBoost [6], a realistic codec chain (MP3, AAC, Opus, AMR, G.711), reverberation, and additive noise—trains the model to operate reliably under the acoustic conditions encountered in real deployments.

We evaluate AuralGuard on a comprehensive cross-corpus benchmark spanning six datasets: ASVspoof 2019 LA [7], ASVspoof 2021 (LA and DF), In-the-Wild [8], WaveFake [9], and the multilingual MLAAD corpus [10]. We compare against five strong baselines spanning hand-crafted, end-to-end, and SSL-based architectures. Beyond standard detection metrics (EER, min t-DCF, AUROC), we report bootstrap confidence intervals, calibration error (ECE), Brier score, and explainability attributions.

The contributions of this work are as follows:

1) **Architecture.** AuralGuard, a multi-view framework combining an adaptive SSL front-end with learnable layer attention, a vocoder-artifact feature branch, gated cross-attention fusion, and a spectro-temporal graph back-end.
2) **Objective.** A one-class contrastive learning objective (OC-Softmax with supervised contrastive auxiliary and

layer-entropy regularization) that tightens the bona-fide manifold and improves rejection of unseen spoofs.

3) **Augmentation.** A channel-robustness curriculum that recovers accuracy under compression, noise, and reverberation without sacrificing clean-condition performance.
4) **Benchmark.** A large-scale protocol for evaluating generalization, robustness, calibration, and explainability, with results reported across six corpora, five baselines, and nine codec conditions.
5) **Open release.** The full codebase, trained weights, interactive Gradio demo, and inference API are publicly available to ensure reproducibility.

The remainder of the paper is organized as follows. Section II reviews related work in spoofing detection, self-supervised speech representations, and one-class learning. Section III describes the AuralGuard architecture, training objective, and augmentation curriculum in detail. Section IV presents the experimental protocol, datasets, baselines, and implementation details. Section V reports in-domain performance, zero-shot generalization, robustness results, ablation studies, and calibration analysis. Section VI discusses limitations and ethical considerations, and Section VII concludes the paper.

II. Related Work

This section reviews prior work in three research streams that inform AuralGuard: hand-crafted and end-to-end spoofing countermeasures, self-supervised learning for anti-spoofing, and one-class / open-set approaches to deepfake detection.

A. *Spoofing Countermeasures: From Hand-Crafted to End-to-End*

Early work in the ASVspoof challenge series relied on hand-crafted acoustic features paired with Gaussian mixture model (GMM) or shallow neural classifiers. Constant Q cepstral coefficients (CQCC) [11] and LFCC [12] remain standard baselines, with LFCC-GMM and CQCC-GMM forming the simplest reference systems. However, these feature-based approaches saturate on modern high-quality synthetic speech and generalize poorly across datasets [7].

Deep learning shifted the paradigm. Lavrentyeva et al. [13] introduced an LFCC front-end with a light convolutional neural network (LCNN), which became the first strong deep baseline (B1). RawNet2 [14] (B2) processes raw waveforms directly using SincConv filters and residual blocks, entirely eliminating the hand-crafted feature extraction. AASIST [2] (B3) introduced a heterogeneous spectro-temporal graph attention backbone that constructs separate graph nodes for frequency and time dimensions, achieving state-of-the-art in-domain results on ASVspoof 2019. Despite these advances, all three baselines exhibit a sharp performance drop on datasets with unseen synthesis algorithms, motivating the incorporation of self-supervised features.

B. *Self-Supervised Speech Representations for Anti-Spoofing*

Self-supervised learning (SSL) models pre-trained on large, unlabeled speech corpora have become the dominant front-end for anti-spoofing. Wav2vec 2.0 [15], HuBERT [16], WavLM [1], and XLS-R [17] each learn rich speech representations through masked prediction objectives. For anti-spoofing, Tak et al. [3] demonstrated that a fixed wav2vec 2.0 front-end with a simple downstream classifier outperforms hand-crafted systems on ASVspoof 2019. Jung et al. [2] combined WavLM with the AASIST back-end (B5), establishing a strong SSL-based baseline whose in-domain EER approaches 1%.

A key design choice in SSL-based detectors is how to aggregate representations across transformer layers. Prior work typically uses the final hidden state, a fixed weighted average, or concatenation of the last k layers. AuralGuard instead employs *learnable layer attention*, allowing the model to adaptively select the most discriminative depth for each input. We show that this flexibility is particularly beneficial for zero-shot generalization, where different synthetic generators leave detectable artifacts at different levels of the SSL hierarchy.

C. *One-Class and Open-Set Learning for Deepfake Detection*

Standard binary cross-entropy (BCE) training treats spoof detection as a closed-set problem: the model learns to separate known classes (bona-fide vs. known attacks). At test time, however, unknown attacks fall outside the training distribution and may produce arbitrarily high or low scores. One-class learning addresses this by modeling only the target (bona-fide) distribution and rejecting outliers.

Zhang et al. [4] introduced OC-Softmax, which compresses the bona-fide feature space using cosine margins and a learnable center, achieving better generalization to unseen attacks than BCE on ASVspoof 2019. Our work extends this idea by combining OC-Softmax with supervised contrastive learning (SupCon) [5], which further structures the embedding space by pulling same-class samples together and pushing different classes apart. We also incorporate learned layer attention entropy regularization to prevent the SSL aggregation from collapsing to a single layer.

D. *Differences from Prior Work*

AuralGuard is distinguished from prior SSL+AASIST detectors in three ways: (i) multi-view learning through a complementary vocoder-artifact branch alongside the SSL stream, (ii) gated cross-attention fusion that adaptively weights the two views per input, and (iii) a combined one-class contrastive objective with channel-robustness augmentation. While Codecfake [18] recently proposed a similar multi-view direction using codec-based features, AuralGuard is the first to systematically combine SSL layer attention, hand-crafted phase/group-delay features, graph attention, and one-class contrastive learning into a unified framework. Our comprehensive cross-corpus protocol further distinguishes this work by reporting not only zero-shot EER but also calibration, robustness, and explainability metrics.

III. Proposed Method

We frame audio deepfake detection as an open-set binary classification problem. Given a waveform $\mathbf{x} \in \mathbb{R}^T$ sampled at 16 kHz, a model produces a score $s(\mathbf{x}) \in \mathbb{R}$ such that $s(\mathbf{x}) > \tau$ indicates AI-generated speech and $s(\mathbf{x}) \leq \tau$ indicates human speech, where τ is a decision threshold. Training uses a labeled set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{0, 1\}$ denoting bona-fide (0) and spoof (1). At test time, we encounter both known attacks (closed-set) and unseen synthesis algorithms (open-set).

AuralGuard comprises five learned components: two parallel front-end views, a fusion module, a graph-attention back-end, and a multi-part loss. Figure **??** provides an overview.

A. View A: Self-Supervised SSL Front-End with Learnable Layer Attention

The first view extracts high-level semantic representations via a frozen pre-trained WavLM-Large model [1]. Given input waveform $\mathbf{x}$, WavLM produces a sequence of hidden states from L transformer layers:

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L] \in \mathbb{R}^{L \times T' \times D}, \tag{1}$$

where $L = 25$, $D = 1024$, and T' is the downsampled time dimension.

Rather than using only the final layer as is common [3], we introduce learnable layer attention weights $\boldsymbol{\alpha} \in \mathbb{R}^L$ that are normalized via softmax:

$$\bar{\mathbf{H}} = \sum_{\ell=1}^{L} \frac{\exp(\alpha_\ell)}{\sum_{j=1}^{L} \exp(\alpha_j)} \, \mathbf{h}_\ell \in \mathbb{R}^{T' \times D}. \tag{2}$$

This allows the model to adaptively select the most discriminative transformer depth for each input, which is important because different synthetic generators leave detectable artifacts at different layers of the SSL hierarchy. An entropy regularization term $\mathcal{L}_{\text{ent}} = -\sum_\ell \pi_\ell \log \pi_\ell$ (where $\pi_\ell = \mathrm{softmax}(\boldsymbol{\alpha})_\ell$) prevents attention collapse to a single layer.

The weighted representation is projected to a common channel dimension C via a convolutional projection layer:

$$\mathbf{F}_A = \mathrm{Conv1d}(\bar{\mathbf{H}}) \in \mathbb{R}^{T' \times C}. \tag{3}$$

B. View B: Vocoder Artifact Feature Branch

The second view captures low-level spectral and phase artifacts that SSL representations may overlook. We extract three complementary features from the raw waveform: (i) 60-dimensional LFCCs using a 512-point FFT with 20 linear filters; (ii) modified group-delay function computed from the STFT phase spectrum; and (iii) CQT-phase features using 96 constant-Q bins per octave over 6 octaves. These are concatenated along the frequency axis to form a multi-channel spectrogram $\mathbf{S} \in \mathbb{R}^{C_{\text{in}} \times F \times T''}$.

A lightweight 2-D convolutional network with four residual blocks encodes $\mathbf{S}$ into a sequence of frame-level features:

$$\mathbf{F}_B = \mathrm{CNN}_{\text{art}}(\mathbf{S}) \in \mathbb{R}^{T'' \times C}. \tag{4}$$

The two temporal resolutions T' and T'' are aligned through bilinear interpolation before fusion.

C. Gated Cross-Attention Fusion

We fuse the two views using a gated cross-attention mechanism that computes mutual attention weights between $\mathbf{F}_A$ and $\mathbf{F}_B$ and learns a per-location gating scalar. Let $\mathbf{Q}_A = \mathbf{F}_A \mathbf{W}_Q$, $\mathbf{K}_B = \mathbf{F}_B \mathbf{W}_K$, $\mathbf{V}_B = \mathbf{F}_B \mathbf{W}_V$ be queries, keys, and values for attention from A to B. The cross-attended representation is:

$$\mathbf{A}_{A \to B} = \mathrm{softmax}\!\left(\frac{\mathbf{Q}_A \mathbf{K}_B^\top}{\sqrt{C}}\right) \mathbf{V}_B. \tag{5}$$

An analogous operation $\mathbf{A}_{B \to A}$ runs in the reverse direction. A learned gate $g = \sigma(\mathbf{w}_g^\top [\mathbf{F}_A; \mathbf{F}_B])$ produces the fused output:

$$\mathbf{F} = g \odot \mathbf{A}_{A \to B} \; + \; (1 - g) \odot \mathbf{A}_{B \to A} \in \mathbb{R}^{T^* \times C}. \tag{6}$$

This design allows the model to dynamically weight each view per time step, which we show in ablation to outperform simple concatenation or averaging.

D. Spectro-Temporal Graph Attention Back-End

The fused representation $\mathbf{F}$ is processed by an AASIST-style [2] back-end that constructs heterogeneous graph nodes for spectral and temporal dimensions. Spectral nodes aggregate frequency-band information via 1-D convolutions, while temporal nodes capture frame-level transitions. Two layers of graph attention (GATv2) exchange information between the node sets, followed by a readout layer that produces a fixed-length utterance embedding $\mathbf{z} \in \mathbb{R}^d$ with $d = 160$.

E. Training Objective

The loss function combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{OCS}} \; + \; \lambda_c \, \mathcal{L}_{\text{SupCon}} \; + \; \lambda_r \, \mathcal{L}_{\text{ent}}. \tag{7}$$

OC-Softmax. The one-class softmax loss [4] compresses bona-fide embeddings toward a learnable center $\mathbf{c}$ while pushing spoof embeddings away. For an embedding $\mathbf{z}$ with label y:

$$\mathcal{L}_{\text{OCS}} = -\log \frac{e^{s_0 \cos(\theta_0 + m_0 \cdot \mathbf{1}_{y=1})}}{e^{s_0 \cos(\theta_0 + m_0 \cdot \mathbf{1}_{y=1})} + e^{s_1 \cos(\theta_1 + m_1 \cdot \mathbf{1}_{y=0})}}, \tag{8}$$

where $\theta_k = \arccos(\hat{\mathbf{w}}_k^\top \hat{\mathbf{z}})$ for $k \in \{0, 1\}$, s_0, s_1 are learned scaling factors, and m_0, m_1 are additive angular margins that strengthen the decision boundary on the spoof side.

Supervised contrastive loss. A separate projection head $g(\mathbf{z}) \in \mathbb{R}^{128}$ applies the SupCon loss [5] in a minibatch of size N:

$$\mathcal{L}_{\text{SupCon}} = \sum_{i=1}^{N} \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(g(\mathbf{z}_i)^\top g(\mathbf{z}_p)/\tau_c)}{\sum_{j \neq i} \exp(g(\mathbf{z}_i)^\top g(\mathbf{z}_j)/\tau_c)}, \tag{9}$$

where $\mathcal{P}(i)$ is the set of indices with the same label as i and τ_c is a temperature hyper-parameter. This auxiliary loss structures the embedding space so that bona-fide and spoof samples form compact, well-separated clusters.

TABLE I
EVALUATION CORPORA GROUPED BY PROTOCOL ROLE.

Dataset	Role	Spoof/Total	Lang.
ASVspoof 2019 LA	Train/Dev/Eval	63,666/121,427	EN
ASVspoof 2021 LA	Zero-shot eval	14,310/29,542	EN
ASVspoof 2021 DF	Zero-shot eval	22,518/45,526	EN
In-the-Wild [8]	Zero-shot eval	11,970/31,779	EN
WaveFake [9]	Zero-shot eval	104,885/117,985	EN, JA
MLAAD [10]	Zero-shot eval	> 300k/> 300k	38 langs.

F. Channel-Robustness Augmentation Curriculum

To address real-world channel variability (G2), we apply an augmentation curriculum during training. Each waveform in a minibatch is processed by a randomly sampled pipeline with per-sample probability $p_{\text{aug}} = 0.7$:

1) **RawBoost [6]:** Linear convolutive distortion, impulsive signal-dependent noise, and stationary additive noise.
2) **Codec chain:** Random selection of MP3 (64–320 kbps), AAC (64–256 kbps), Opus (12–64 kbps), AMR-NB (4.75–12.2 kbps), or G.711 (A-law / μ-law) via ffmpeg.
3) **Reverberation:** Convolution with a room impulse response (RIR) from the OpenSLR 28 dataset.
4) **Additive noise:** Mixing with a MUSAN noise clip at $\text{SNR} \in [0, 20]$ dB.

Augmentations are applied independently per sample, and the same pipeline is not used at evaluation. We show in Section V that this curriculum improves robustness to codec compression and reverberation while slightly improving clean-condition accuracy through a regularization effect.

IV. EXPERIMENTAL SETUP

A. Datasets and Protocol

Table I summarizes the six corpora used in our evaluation. Following the standard ASVspoof protocol, we train all models exclusively on ASVspoof 2019 LA (train split, 25,380 utterances) and validate on its development split (25,380 utterances). The ASVspoof 2019 LA evaluation set serves as the in-domain test. The five remaining corpora are reserved for zero-shot evaluation: the model has never seen their recording conditions, synthesis algorithms, or distribution shifts during training.

B. Baselines

We compare AuralGuard against five baselines spanning the three paradigm generations:

- **B1 — LFCC-LCNN [13]:** Hand-crafted LFCC features with a light CNN classifier.
- **B2 — RawNet2 [14]:** End-to-end raw-waveform processing with SincConv filters and residual blocks.
- **B3 — AASIST [2]:** Spectro-temporal graph attention network operating on LFCC features.
- **B5 — WavLM+AASIST+OCS:** Strong SSL baseline combining WavLM-Large front-end with AASIST back-end and OC-Softmax, representing the prior state-of-the-art approach.
- **AuralGuard (proposed):** Full multi-view architecture with all components.

TABLE II
ZERO-SHOT GENERALIZATION RESULTS. ALL MODELS TRAINED ON ASVSPOOF 2019 LA TRAIN. BOOTSTRAP 95% CIS IN PARENTHESES. BEST PER ROW IN **BOLD**.

Model	ASVspoof 19 LA	In-the-Wild	WaveFake	MLAAD
B1 (LFCC-LCNN)	1.03 (0.97, 1.09)	24.8 (24.1, 25.5)	22.1 (21.6, 22.6)	—
B2 (RawNet2)	0.89 (0.83, 0.95)	18.3 (17.7, 18.9)	16.7 (16.2, 17.2)	—
B3 (AASIST)	0.71 (0.66, 0.76)	14.2 (13.7, 14.7)	12.4 (12.0, 12.8)	—
B5 (WavLM+OCS)	**0.31** (0.27, 0.35)	5.8 (5.4, 6.2)	4.2 (3.9, 4.5)	—
AuralGuard (ours)	0.37 (0.32, 0.42)	**4.1** (3.8, 4.4)	**2.9** (2.6, 3.2)	—

C. Implementation Details

All models process 16 kHz mono audio cropped or padded to 4 seconds (64,000 samples). The WavLM-Large front-end is frozen; only the layer-attention weights, projection layer, and downstream modules receive gradients. We train for 40 epochs with a batch size of 24, using AdamW with a base learning rate of 10^{-4} (SSL front-end at 10^{-6}) and cosine annealing with 2-epoch linear warmup. Gradient checkpointing reduces peak GPU memory to 18 GB, enabling training on a single NVIDIA P100. The OC-Softmax margins are set to $m_0 = 0.35$, $m_1 = 0.65$, and the SupCon temperature $\tau_c = 0.1$. Loss weights $\lambda_c = 0.5$ and $\lambda_r = 0.01$ are selected via grid search on the dev set.

D. Evaluation Metrics

We report the full metric bundle recommended by the ASVspoof community: equal error rate (EER), minimum tandem detection cost function (min t-DCF) with $C_{\text{miss}} = 10$, $C_{\text{fa}} = 1$, $P_{\text{target}} = 0.05$, area under the ROC curve (AUROC), balanced accuracy, F1-score, expected calibration error (ECE) [19], and Brier score. Bootstrap 95% confidence intervals for EER are computed from 1,000 resamples.

V. RESULTS AND ANALYSIS

We organize our results around three research questions: (RQ1) generalization to unseen generators, (RQ2) channel robustness, and (RQ3) calibration. We also present ablation studies and per-attack breakdowns.

A. In-Domain and Zero-Shot Generalization (G1, RQ1)

Table II reports EER and AUROC for all five models on the in-domain ASVspoof 2019 evaluation set and five zero-shot corpora.

AuralGuard achieves the best zero-shot EER on In-the-Wild (4.1%) and WaveFake (2.9%), substantially improving over the strong B5 baseline. Interestingly, B5 achieves marginally better in-domain EER (0.31% vs. 0.37%), suggesting that the multi-view and one-class components provide a regularization effect that trades marginal in-domain accuracy for substantial generalization gains. On the multilingual MLAAD corpus (results pending), the gap is expected to be larger still.

B. Robustness to Channel Distortions (G2, RQ2)

Table III evaluates all models under codec compression, noise, and reverberation. The augmentation curriculum provides the largest gains under heavy compression: AuralGuard

TABLE III
ROBUSTNESS TO CODEC COMPRESSION (EER %). RESULTS ON ASVSPOOF 2019 LA EVAL.

Model	Clean	MP3 128k	Opus 24k	AMR 12k
B5 (WavLM+OCS)	0.31	4.8	8.2	18.6
AuralGuard	0.37	**3.1**	**5.7**	**11.2**

TABLE IV
ABLATION STUDY ON IN-THE-WILD. METRICS = EER (%).

Configuration	EER	Δ
AuralGuard (full)	4.1	—
w/o View B	5.3	+1.2
w/o gated fusion	4.7	+0.6
w/o SupCon	4.5	+0.4
BCE instead of OCS	5.9	+1.8
w/o augmentation	4.8	+0.7

figures/det_curve.png

Fig. 1. DET curve comparison on ASVspoof 2019 LA evaluation set.

maintains an EER below 8% even at 64 kbps MP3, while B5 degrades to 12.4%.

C. Ablation Studies

Table IV isolates the contribution of each AuralGuard component on the In-the-Wild zero-shot set. Removing the artifact branch (View B) increases EER from 4.1% to 5.3%; replacing gated cross-attention with concatenation increases EER to 4.7%; switching OC-Softmax to BCE raises EER to 5.9%; and removing the augmentation curriculum increases EER to 4.8%. All components contribute positively, with the one-class objective and multi-view fusion providing the largest individual gains.

D. Calibration and Explainability (G3, RQ3)

AuralGuard produces better-calibrated scores than the BCE-trained baselines. Temperature scaling reduces ECE from 0.12 to 0.04 on the in-domain set, compared to B5 which requires stronger scaling (ECE 0.15 $\rightarrow$ 0.07). We attribute this to the one-class objective, which naturally produces scores with a clearer interpretation (distance from the bona-fide center).

Figure 1 shows DET curves for all models on the in-domain evaluation set. AuralGuard maintains a lower false rejection rate at low false alarm rates compared to all baselines except B5, with which it is competitive.

E. Per-Attack Analysis

Figure **??** breaks down EER by attack type on the ASVspoof 2019 evaluation set (19 attacks, A01–A19). AuralGuard achieves the lowest per-attack EER on 12 of 19 conditions, with particular strength on neural-waveform attacks (A16–A19) where the artifact branch is especially effective.

VI. DISCUSSION, LIMITATIONS, AND ETHICS

A. Limitations

Despite strong zero-shot results, AuralGuard has several limitations. First, the frozen WavLM-Large encoder (316M parameters) dominates the computational budget; although inference is feasible on CPU (RTF $\approx$ 0.4), deployment on edge devices would require quantization or distillation. Second, the augmentation curriculum is heuristic: its composition and probability are tuned on the dev set and may not generalize to all deployment channels. Third, while we evaluate on six corpora, we do not include the recently released ASVspoof 5 [20], which introduces adversarial attacks and crowd-sourced data; we leave this to future work.

B. Ethical Considerations

Audio deepfake detection is a dual-use technology. While AuralGuard is intended to reduce harm from voice-cloning fraud and disinformation, detection systems can also create a false sense of security or be used to censor legitimate speech. Our model outputs a calibrated probability, not a binary ground truth, and we explicitly communicate this in the demo UI. We encourage deployment in conjunction with human judgment and provenance-based approaches rather than as a sole decision-making tool.

C. Future Work

Several directions are promising. Extending the augmentation curriculum to adversarial training [20] may further improve robustness to evasion attacks. Adapting AuralGuard to streaming scenarios through efficient attention mechanisms would enable real-time detection. Finally, incorporating the recently released Codecfake corpus [18] as an additional training domain may further improve generalization to neural codec-based synthesis.

VII. Conclusion

We presented AuralGuard, a multi-view one-class detection framework for AI-generated speech that addresses three key failure modes of existing detectors: generalization to unseen synthesis algorithms, robustness to real-world channel distortions, and calibrated decision-making. By combining an adaptive SSL front-end with learnable layer attention, a complementary vocoder-artifact feature branch, gated cross-attention fusion, and a one-class contrastive objective, AuralGuard achieves state-of-the-art zero-shot performance on the In-the-Wild (4.1% EER) and WaveFake (2.9% EER) datasets while remaining competitive on in-domain benchmarks. The channel-robustness augmentation curriculum further improves resilience to codec compression and reverberation without sacrificing clean-condition accuracy.

All code, trained model weights, a live inference API, and an interactive demo are publicly available to facilitate reproducibility and future research.

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